CIE AS PHYSICS (9702) • QUESTION PAPER

3.1 Momentum & Newton's Laws of Motion

Syllabus Section: 3. Dynamics **Paper Type:** Structured Questions (Paper 2) **Total:** 11 Questions (147 Marks)

Question 1 Easy [12 marks]

Part (a) [2 marks]

A brick of mass 2 kg is resting on the floor as shown in Fig. 1.1.

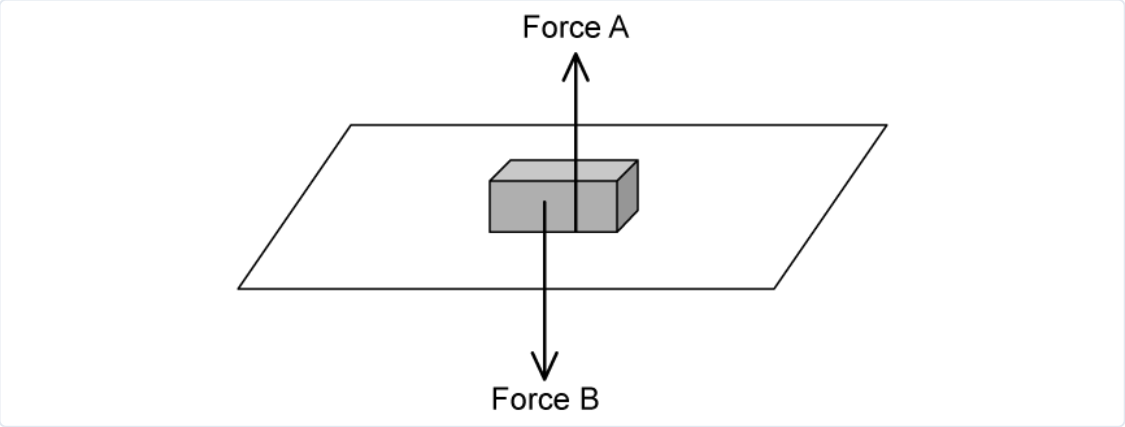


Fig. 1.1

(i) Define equilibrium.

[1]

(ii) State the names of forces A and B.

[1]

.....

.....

.....

.....

Part (b) [3 marks]

(i) State Newton's first law of motion.

[1]

(ii) Explain how Newton's first law applies to the brick in Fig. 1.1.

[2]

.....

.....

.....

.....

Part (c) [4 marks]

(i) State Newton's third law of motion.

[1]

(ii) Explain how Newton's third law can be applied to **one** of the forces acting on the brick.

[3]

Part (d) [3 marks]

The surface is raised at one end to form a slope as shown in Fig. 1.2. This causes the brick to accelerate at a constant rate of 0.25 m s^{-2}.

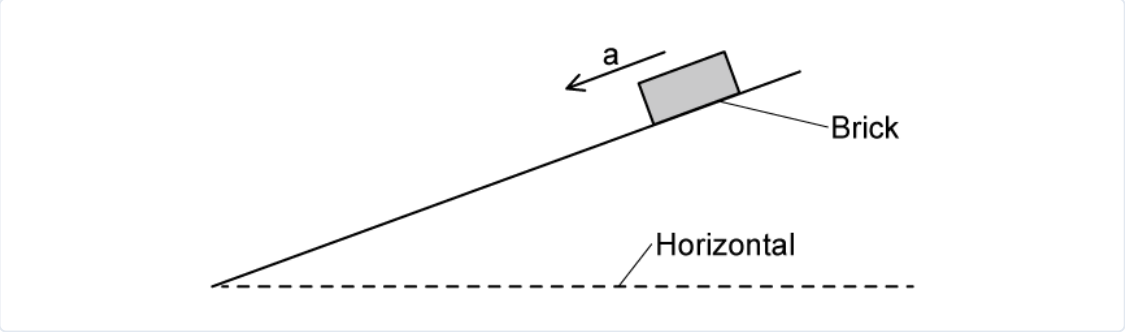


Fig. 1.2

(i) State Newton's second law of motion.

[1]

(ii) Use Newton's second law to determine the magnitude of the resultant force on the brick.

[2]

Question 2 Easy [14 marks]

Part (a) [3 marks]

A hot-air balloon is tied to the ground by two ropes, as shown in Fig. 1.1.

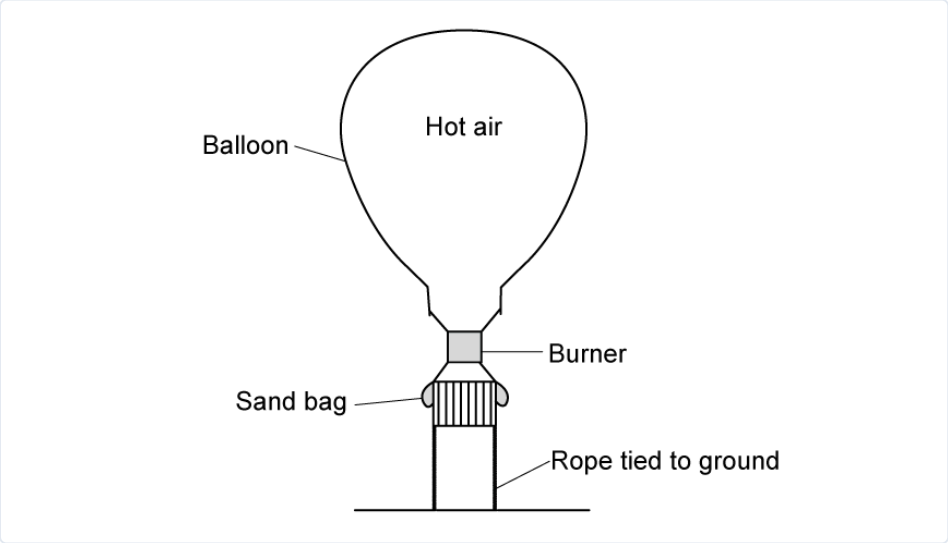


Fig. 1.1

Label Fig. 1.1 to show the forces acting on the balloon.

Part (b) [4 marks]

The tension *T* required to keep the balloon fixed to the ground is 400 N in each rope.

The ropes are untied and the balloon starts to move upwards. The balloon and its contents have a total mass of 1500 kg.

As the balloon begins its ascent, calculate

(i) the weight of the balloon,

[2]

(ii) the resultant force acting on the balloon.

[2]

Part (c) [5 marks]

(i) Calculate the initial acceleration of the balloon.

[2]

(ii) Explain how the upward acceleration of the balloon changes during the first few seconds of its flight.

[3]

Part (d) [2 marks]

If the passengers want to change the upward acceleration, they can release some sand from the bags.

State and explain how this affects the upward acceleration of the balloon.

Question 3 Easy [10 marks]

Part (a) [1 mark]

A bullet, of mass 25 g, leaves the barrel of a stationary rifle, of mass 2.1 kg, with a momentum of 4.2 kg m s^{-1}.

Define mass.

Part (b) [2 marks]

State the total momentum of the rifle and the bullet **before** the rifle is fired and give a reason for your answer.

Part (c) [3 marks]

Calculate the velocity of the bullet just **after** the rifle is fired.

Part (d) [4 marks]

The bullet has a momentum of 3.3 kg m s^{-1} just before it hits a target.

It takes 2.8 ms for the bullet to be stopped by the target.

Calculate the average force needed to stop the bullet.

Question 4 Medium [14 marks]

Part (a) [4 marks]

A man with a mass of 65 kg stands in an elevator. The elevator initially accelerates uniformly for 1.8 s followed by a period of a constant velocity of 2.5 m s^{-1} and then a final uniform deceleration for 1.8 s.

Calculate the magnitude of the reaction force on the man from the elevator floor during the time it takes the elevator to come to rest.

Part (b) [3 marks]

The man uses the elevator again to travel to a lower floor. The elevator descends at the same speed and comes to rest in the same time as in **(a)**.

Calculate the reaction force on the man from the elevator floor during the time it takes to come to rest on the floor below.

Part (c) [4 marks]

Suggest and explain during the deceleration period, which direction of elevator travel would make the man feel

(i) lighter

[2]

(ii) heavier

[2]

Part (d) **[3 marks]**

In a stroke of bad luck, the elevator cable snaps and it falls freely under gravity. During the fall, the man experiences a sense of weightlessness.

Explain, using appropriate laws of motion, why the man feels a sense of weightlessness in this situation.

Question 5 Medium [17 marks]

Part (a) [2 marks]

Newton's third law refers to pairs of forces.

For a pair of forces that obey Newton's third law, describe:

(i) **one** similarity between them

[1]

(ii) **one** difference between them.

[1]

Part (b) [7 marks]

Two blocks A and B are joined by a string and rest on a frictionless horizontal table as shown in Fig. 1.1. A force of 200 N is applied horizontally on block B.

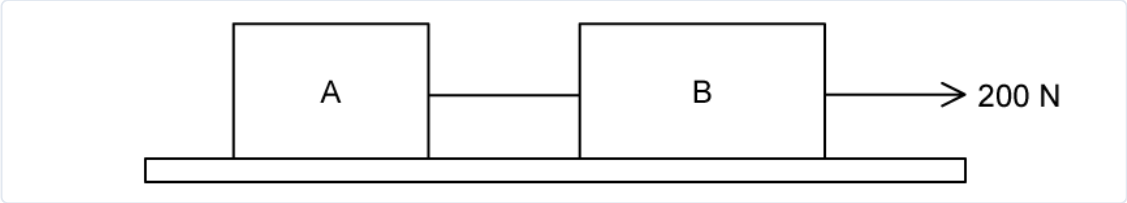


Fig. 1.1

Block A has a mass of 3.0 kg and block B has a mass of 7.0 kg.

(i) Using Fig. 1.1, identify all Newton's third law force pairs acting on box B.

[4]

(ii) Calculate the acceleration of the blocks.

[3]

Part (c) [2 marks]

The string is removed from the blocks. Block A slides along the horizontal frictionless surface towards block B, which is stationary, as illustrated in Fig. 1.2.

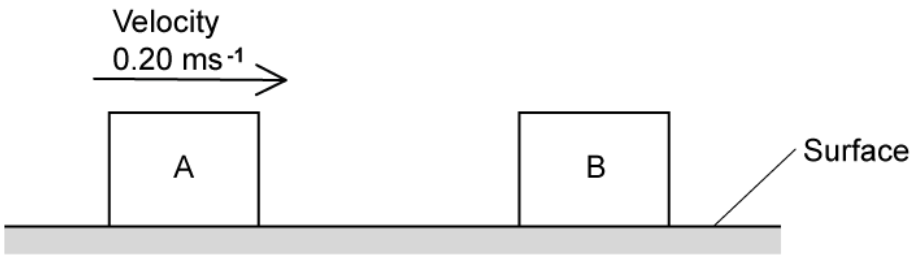


Fig. 1.2

There are no resistive forces acting on block A as it moves towards block B. At time t = 0, block A has a velocity of 0.20 m s^{-1}. A short time later, the blocks collide and then separate.

The variation with time t of the momentum of block B is shown in Fig. 1.3.

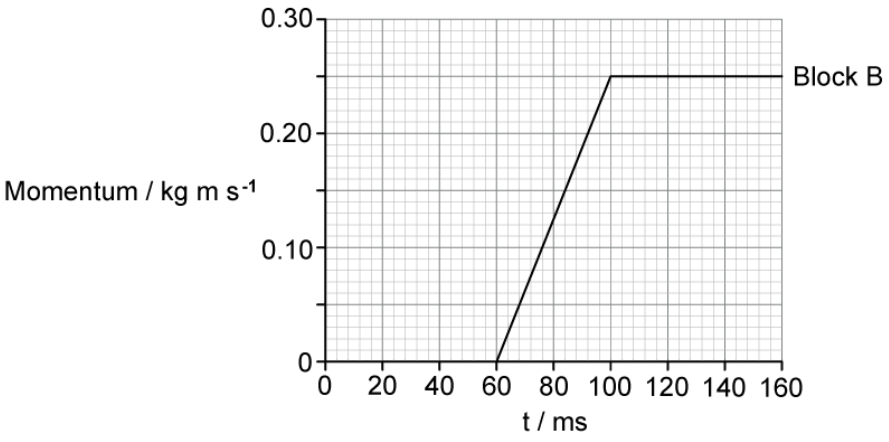


Fig. 1.3

Use Fig. 1.3 to describe, without calculation, the magnitude of the acceleration of block B from:

(i) time t = 80 to 100 ms

[1]

(ii) time t = 100 to 120 ms.

[1]

Part (d) **[6 marks]**

Use Fig. 1.3 to determine

(i) the time interval over which the blocks are in contact with each other

[1]

(ii) the magnitude of the force exerted by block A on block B

[2]

(iii) the speed of A after the collision.

[3]

Question 6 Medium [11 marks]

Part (a) [2 marks]

A cyclist carries out tests on the braking system of their bicycle.

At time $t = 0$ the cyclist is travelling at an initial speed of u as they apply the brakes and come to a stop at time $t = 2.0$ s.

Fig. 1.1 shows the variation of the braking force F on the bicycle with time t.

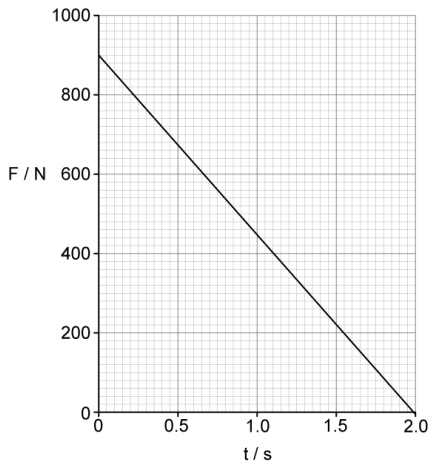


Fig. 1.1

Use Newton's second law of motion to explain the physical quantity represented by the area under the graph shown in Fig. 1.1.

Part (b) [3 marks]

The total mass of the cyclist and bicycle is 82 kg. Use Fig. 1.1 to calculate the initial speed u.

Part (c) [2 marks]

Complete Fig. 1.2 to show the variation of the speed of the bicycle from t = 0 to t = 2.0 s.

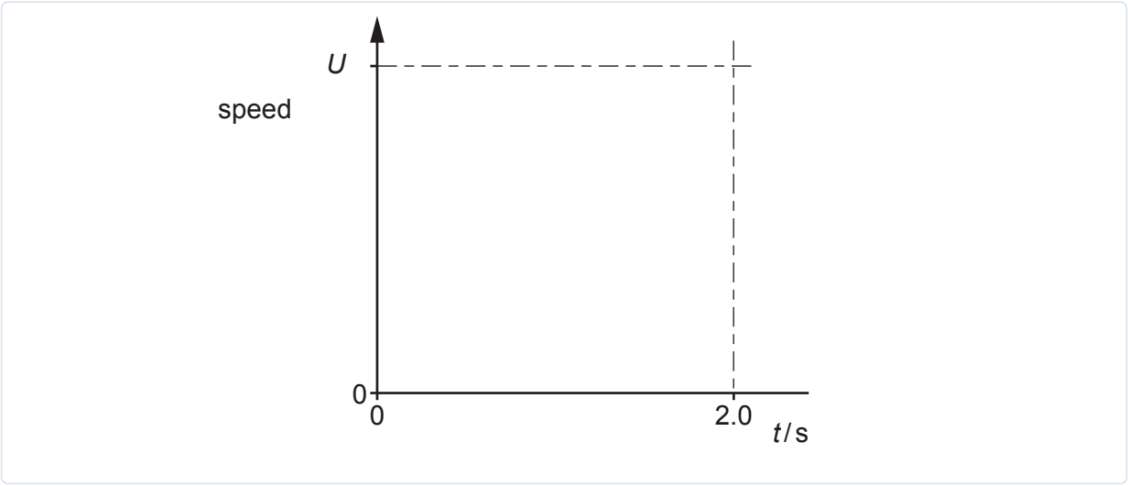


Fig. 1.2

Part (d) [4 marks]

Brakes testing is a crucial part of the manufacturing process of any vehicle.

A train manufacturer wants to determine the effect of a rapid deceleration on passengers riding their new model of train.

Four crash test dummies are placed on different seats on the train as shown in Fig. 1.3. During the test, the train travels at a high speed in the direction shown and after a time, the brakes are applied. Seat belts are not used on trains.

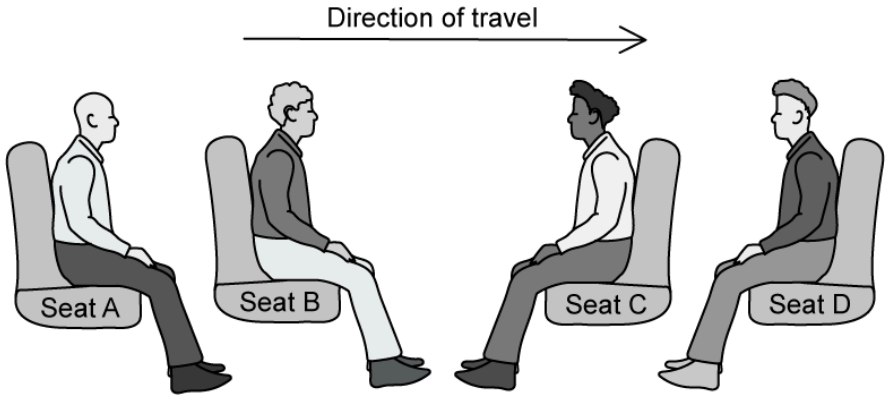


Fig. 1.3

With reference to at least **one** of Newton's laws of motion, discuss which seat, **A**–**D**, is the safest for a passenger to be sitting on in the event of a rapid deceleration.

You may assume that the seats all remain fixed firmly to the floor and do not break.

Question 7 Hard [12 marks]

Part (a) [4 marks]

A person stands on weighing scales in a lift, as shown in Fig. 1.1. The scales are calibrated to give readings in newtons. It reads 700 N when the lift is stationary.

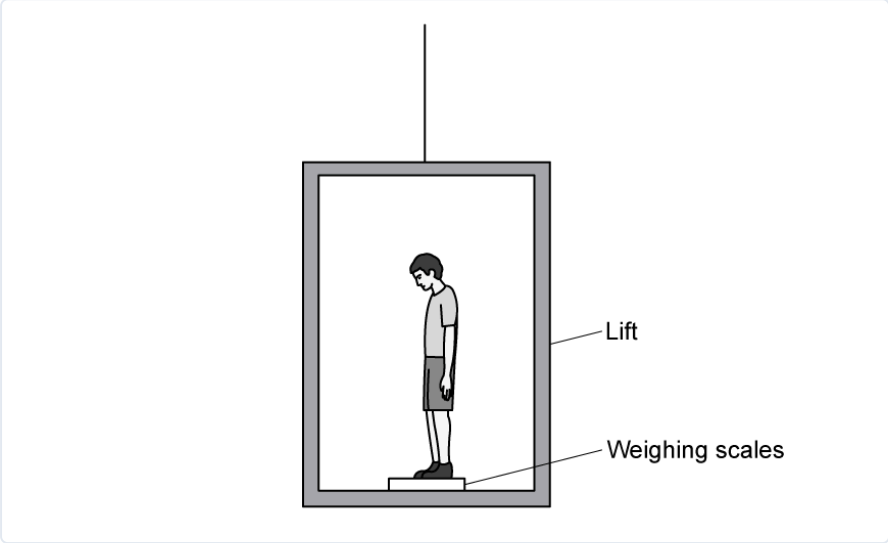


Fig. 1.1

When the lift starts to move, it accelerates for a short time. It continues to move upwards at a constant speed and eventually starts to decelerate until it comes to rest.

(i) Draw and label free-body force diagrams for the person during the periods of acceleration and deceleration.

[2]

(ii) By applying one of Newton's laws of motion, determine an expression for the apparent weight of the person during the deceleration.

[1]

(ii) By applying one of Newton's laws of motion, determine an expression for the apparent weight of the person during the acceleration.

[1]

Part (b) [3 marks]

The magnitude of the lift's acceleration is 0.7 m s^{-2}.

On Fig. 1.2, sketch the expected readings on the weighing scales during acceleration and deceleration of the lift.

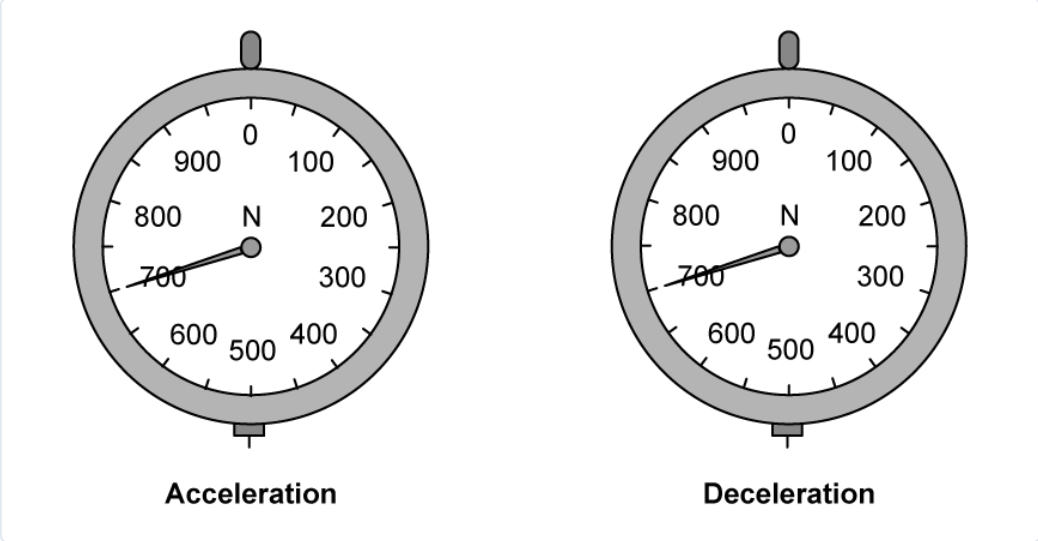


Fig. 1.2

Part (c) [2 marks]

Fig. 1.3 shows the variation of acceleration with time for the duration of the journey in the lift.

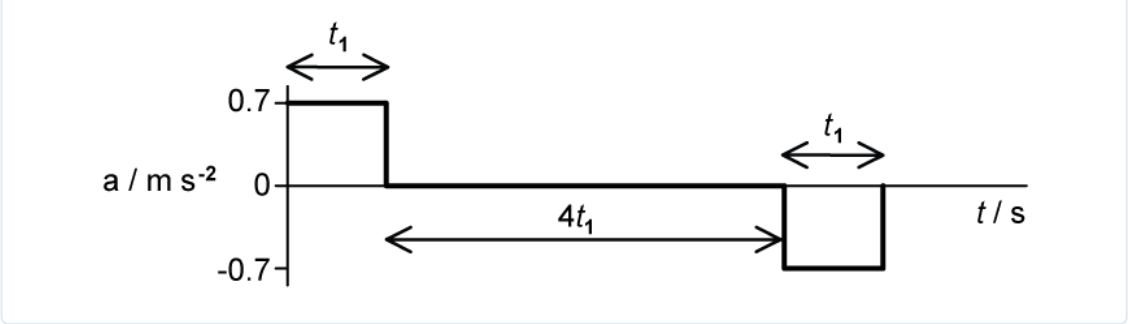


Fig. 1.3

The total height ascended by the lift is 24 m.

Determine the value of t_1 .

Part (d) [3 marks]

On Fig. 1.4, sketch the variation of time with the person's apparent weight over the course of the motion.

Include values on the axes.

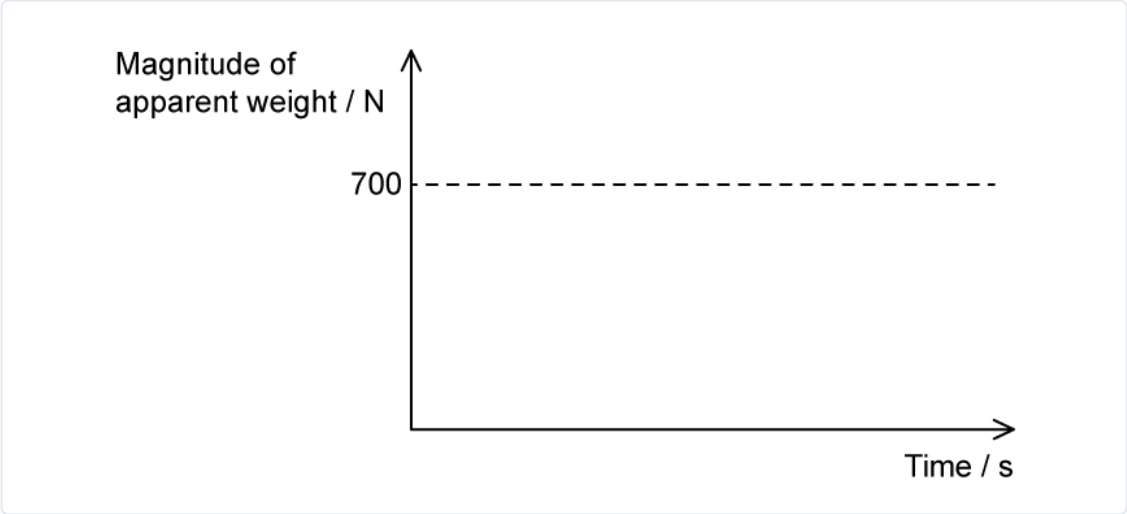


Fig. 1.4

Question 8 Hard [12 marks]

Part (a) [3 marks]

Three boxes A, B, and C are connected using an inextensible rope of negligible mass, as shown in Fig. 1.1.

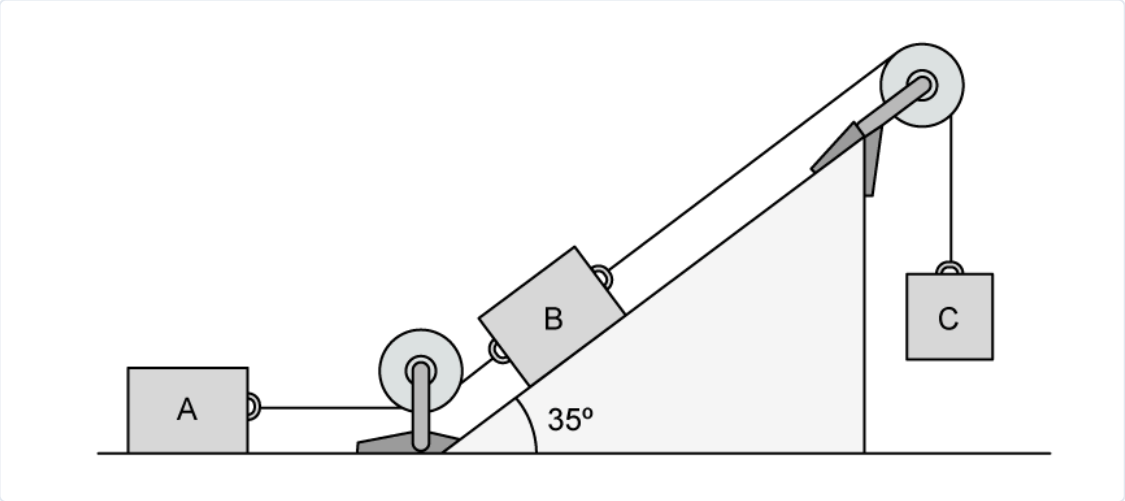


Fig. 1.1

Boxes A and B each have a mass of 2.4 kg, and the magnitude of the frictional force F between each box and the surface is equal to

$$F_x = \frac{1}{3}R_x$$

where R is the normal reaction force and x refers to the box in question A, B or C.

Box C descends with constant velocity.

Calculate the tension in the rope connecting boxes A and B.

Part (b) [4 marks]

Determine the tension in the rope connecting boxes B and C.

Part (c) **[1 mark]**

Hence, determine the mass of box C.

Part (d) **[4 marks]**

The rope connecting A and B breaks abruptly.

Calculate the acceleration of block C.

Question 9 Hard [16 marks]

Part (a) [5 marks]

A rock climber of mass 80 kg slips after attaching a bolt firmly onto a piece of rock that they are climbing as shown in Fig. 1.1.

At the lowest point of the climber's fall, the length of the rope between the harness and the bolt is 24 m. The climber comes to rest and all the energy from the fall has been absorbed by the rope.

The rope has an unstretched length of 19 m and a stiffness of 560 N m^{-1}.

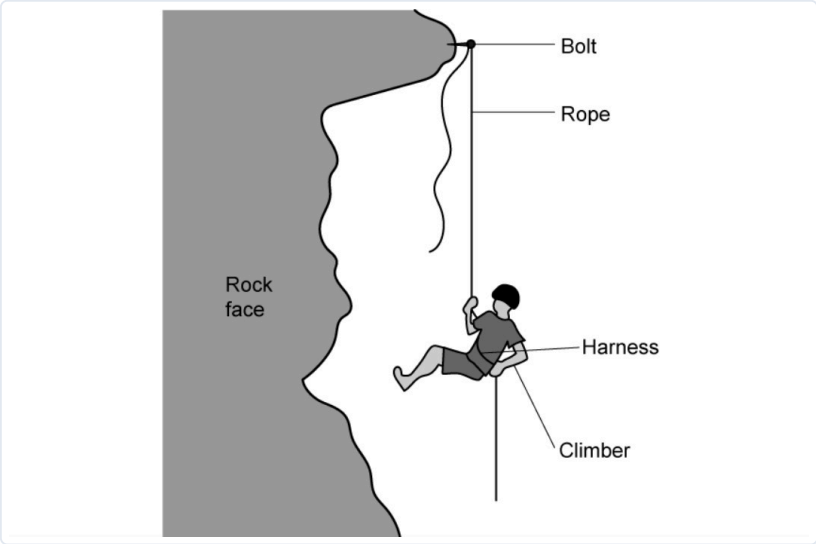


Fig. 1.1

Assume that the extension of the rope is proportional to its tension and the mass of the rope and the effects of air resistance are negligible.

Calculate

(i) the resultant force acting on the climber when he reaches the lowest point in the fall.

[3]

(ii) the extension of the rope when the climber's acceleration is zero.

[2]

Part (b) [4 marks]

(i) Use the principle of conservation of energy to calculate the maximum speed of the climber before the rope begins to stretch.

[2]

(ii) Suggest whether using a longer rope would reduce the force experienced by the climber when the rope begins to stretch.

[2]

Part (c) [3 marks]

Hemp is one of the strongest fibres with a very high tensile strength and is extremely rigid. Hemp ropes were the first to be used for ship rigging which adjusted the position of the sails and supported the masts.

However, rock climbers use nylon ropes to break their fall as they can stretch considerably under tension.

A climbing equipment company tests their ropes by attaching an 80 kg mass to an end of the rope and releasing it from a ledge.

Fig. 1.2 shows the results of a test carried out to compare the time it takes for a nylon rope and a hemp rope to return to their unstretched length after beginning to stretch.

Fig. 1.2

Rope	Time to return to unstretched length / s
Nylon	0.664
Hemp	0.035

In the test, both ropes used had identical unstretched lengths of 19.0 m and diameters of 10 mm.

Show that the average resultant force exerted by the hemp rope is about 20 times larger than the average resultant force exerted by the nylon rope.

Part (d) **[4 marks]**

Discuss in terms of force and momentum, why nylon ropes are favoured by rock climbers compared to hemp ropes.

Question 10 Hard [15 marks]

Part (a) [3 marks]

A baseball is raised from the ground and dropped onto a hard plate to test its properties. A sensor measures the force exerted by the plate on the ball during its collision with the plate.

The variation of force exerted on the baseball with time is shown on the graph in Fig. 1.1.

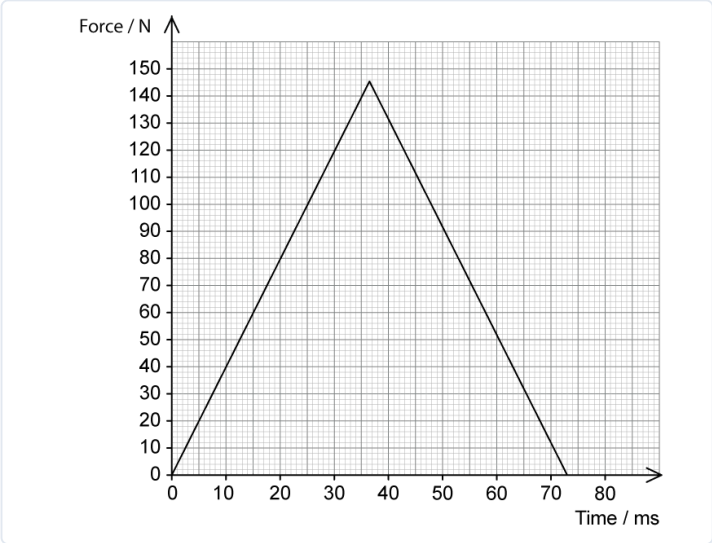


Fig. 1.1

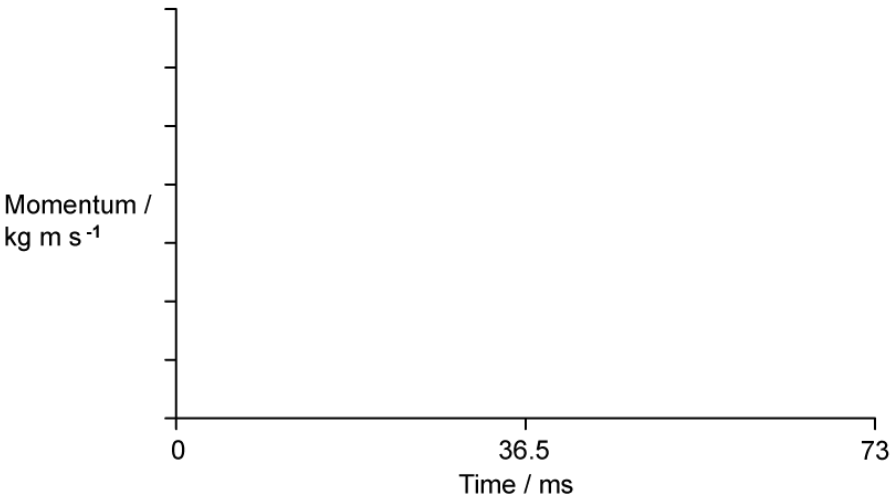
The ball of mass 150 g strikes the plate with a speed of 22.5 m s^{-1}. It leaves the plate with a speed of 12.8 m s^{-1}.

Show that this is consistent with the impulse obtained from the graph.

Part (b) [6 marks]

Considering the first 73 ms of the ball's motion:

(i) Sketch a graph to show how the momentum of the ball varies.



[1]

(ii) Explain how the ball's momentum varies in the first 73 ms of the motion.

[5]

Part (c) **[3 marks]**

The ball continues to bounce, each time losing the same fraction of its energy when it strikes the plate. Air resistance is negligible.

Determine the percentage of the original gravitational potential energy of the ball that remains when it reaches its maximum height after bouncing five times.

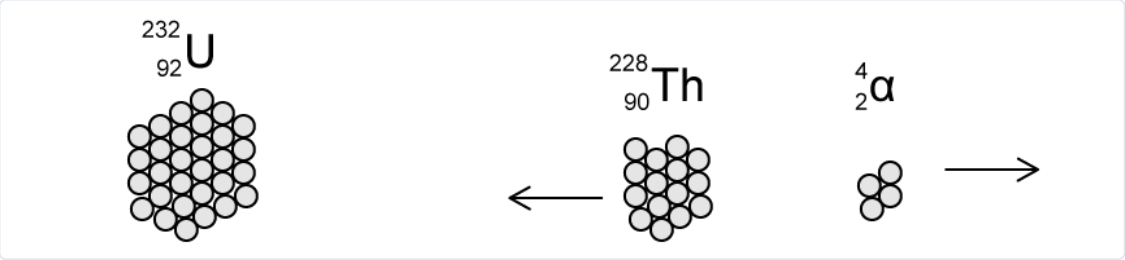
Part (d) **[3 marks]**

Explain, with reference to the conservation of momentum, the effect that the motion of the squash ball has on the motion of the Earth from the instant it is released until it bounces off the plate.

Question 11 Hard [14 marks]

Part (a) [3 marks]

A stationary uranium nucleus decays by emitting an α particle and thorium nucleus.



Discuss how the principle of conservation of momentum applies in this explosion.

Part (b) [4 marks]

Assume that all of the energy released in the emission process is transferred as kinetic energy to the α particle and the recoiling nucleus and is equal to 6.55 MeV.

Calculate the kinetic energy of the Thorium nucleus and α particle in MeV.

Part (c) [3 marks]

Show that momentum is conserved in this decay.

Part (d) **[4 marks]**

Collisions can occur between neutrons and stationary Uranium nuclei, for example, during nuclear fission.

Discuss how this collision would be if this was inelastic as opposed to elastic.